# Joint Estimation of Satellite Attitude and Size Based on ISAR Image Interpretation and Parametric Optimization

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***Abstract*— This article presents a novel approach for the joint estimation of satellite attitude and size based on inverse synthetic aperture radar (ISAR) image interpretation and parametric optimization. The satellite's solar panel, which is segmented from the ISAR image by employing pix2pix generative adversarial network (Pix2pixGAN), is chosen for investigation in this article due to its unique rectangular structure. We innovatively use the principal component analysis (PCA) to explore the satellite solar panel's structural features in an ISAR imagery. The projection matrix is then established to link the extracted features and the satellite's absolute attitude and size. Parametric optimization is established based on the relationship between the extracted features and the satellite's absolute attitude and size. A Broyden–Fletcher–Goldfarb–Shanno (BFGS)-based fast iterative search algorithm is employed to search the satellite's absolute attitude and size simultaneously through an iterative approach. The simulation data are generated from actual satellite orbital parameters and computer-aided-design (CAD) models of the Aqua satellite in the experiments. Simulation experiments verify the effectiveness of the proposed method.**

***Index Terms*— Attitude estimation, geometry characteristics, inverse synthetic aperture radar (ISAR), pix2pix generative adversarial network (Pix2pixGAN), principal component analysis (PCA) parametric optimization, satellite targets.**

## I. Introduction

With the increasing exploration of the space field, more and more satellites are being launched into space [1], [2]. As a result, the acquisition of satellite attitude is essential for analyzing satellite activities, military activities, status, and so on [3]–[7]. On the other hand, the physical size of the major satellite components (e.g., satellite payload size, solar panel length, and width) also plays an important role in satellite cataloging, identification, and classification. Since the inverse synthetic aperture radar (ISAR) can acquire high-resolution 2-D images of satellite targets at long distances and all-day [8]–[10], it has always been chosen to perform the critical task of satellite attitude and size estimation. As the ISAR image of the satellite targets contains rich structural features, it is feasible to estimate the satellite's attitude and size from the ISAR image based on these features and target motion characteristics [11].

Reviewing the conventional satellite attitude and size estimation methods, they can be classified into two basic categories [6], [12]–[15]. One basic approach requires a pre-existing simulation database obtained from a full-angle observation of the target's electromagnetic (EM) simulation model [6], [12]. The measured data, including radar cross section (RCS), high-resolution range profile (HRRP), and ISAR images of the target, are then matched with the preexisting database for searching the target's attitude. This method requires a realistic EM simulation model, which is feasible on some tasks with cooperative targets but fails when the target is uncooperative. Another way is to obtain the ISAR image sequence of the target based on continuous radar observations [13]–[15]. By extracting and associating the target's scatterers from those images, the target's 3-D structure can be obtained based on some 3-D reconstruction methods, such as the factorization-based reconstruction method. The reconstruction results can be used to estimate the satellite's attitude and size. For this approach, the association of scatterers in different ISAR images is not an easy task. Besides, due to the unknown rotation between the acquired 3-D construction and the actual target structure, it is impossible to obtain the target attitude for unknown targets.

Due to the shortcomings and limited applicability of these two traditional approaches, several researchers have innovatively proposed more robust and widely applicable methods in recent years [11], [16], [17]. Zhou *et al.* [11], [18] innovatively used the typical linear features of satellite rectangular solar panels. An optimization function is developed for the satellite's attitude and size estimation by associating the linear structures in different ISAR images. It is a wise way to accomplish this task. After all, almost all satellites are equipped with rectangular solar panels, so this method can be applied to

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almost any satellite target, whether it is cooperative or noncooperative. The disadvantages of this approach are that it requires manual involvement to extract linear features. In addition, the attitude and size of the satellite are estimated separately, which leads to error accumulation and makes the error increase. In [16], the quadratic spatial-variant phase error generated by the target's rotation is considered helpful information. The author considers the satellite solar panel as a plane and compensates for the phase error while estimating the quadratic phase's coefficients, from which the solar panel's normal vector is determined. This characterizes the attitude only up to a rotation about the normal vector. This method is also intelligent in treating quadratic phase errors as valuable information. The attitude estimation can be accomplished with only one ISAR image, which is also applicable when satellite data are insufficient. However, this method can only estimate the solar panel's normal vector but not the orientations of its long and short edges, and the size cannot be estimated. In practice, the coefficient estimation of the quadratic phase is sensitive to noise.

Inspired by [11], this article proposes a more robust method for the joint estimation of satellite attitude and size. The proposed method also takes advantage of the rectangular structural characteristics of the solar panel. Unlike [11], we exploit more robust point features on the solar panel, which are much easier to be extracted than linear features and require no manual involvement. On the other hand, the proposed method treats the solar panel as a unit instead of treating the long and short edges separately as in the [11]. The proposed method also achieves a joint estimation of satellite attitude and size by establishing an optimization function, which reduces the error accumulation of stepwise estimation. This article's core idea is to treat the solar panel as a rectangular unit, segmented from the ISAR image using the pix2pix generative adversarial networks (Pix2PixGANs) [18]. Then, it determines the coordinate set of the solar panel points using the peak extraction technique [19]. The principal component analysis (PCA) [20] method is employed to extract the principal component directions of the point set. Two principal component vectors and eigenvalues of the solar panel are obtained. Based on these vectors and eigenvalues, an optimization function is established to estimate the satellite attitude and size, and the optimal solution is achieved iteratively by an iterative gradient algorithm. The proposed algorithm has the following innovations and advantages compared to the approach in [11].

1) The proposed method is based on scattering point features, which are more robust and easier to acquire than linear features. The PCA is employed to extract eigenvectors and eigenvalues of solar panels, making the whole feature extraction process completely automatable and does not require manual interaction.
2) The proposed method treats the solar panel as a whole. By taking advantage of its rectangular structure and employing ISAR imaging's projection principle, all the parameters, including the long and short edges' orientations and lengths, can be estimated simultaneously. Compared with the approach proposed in [11], the proposed method can simultaneously estimate both the attitude and size, reducing the accumulated errors caused by stepwise estimation.
3) In this article, a parametric optimization function is established, and the solar panel attitude and size parameters can be estimated jointly. By deriving the optimization function gradient, these parameters can be obtained by a Broyden–Fletcher–Goldfarb–Shanno (BFGS)-based quasi-Newton iterative method [21], [22]. Compared to the particle swarm optimization (PSO) algorithm, it dramatically reduces the algorithm's execution time and has a higher estimation accuracy.

The remainder of the article is organized as follows. In Section II, the ISAR image projection model and the characteristics of the PCA process for the solar panel are introduced. In Section III, the implementation details of the proposed algorithm are provided. In Section IV, some experiments are given, including the attitude and size estimation of Aqua satellite and the comparison with some existing methods. In Section V, some conclusions are drawn.

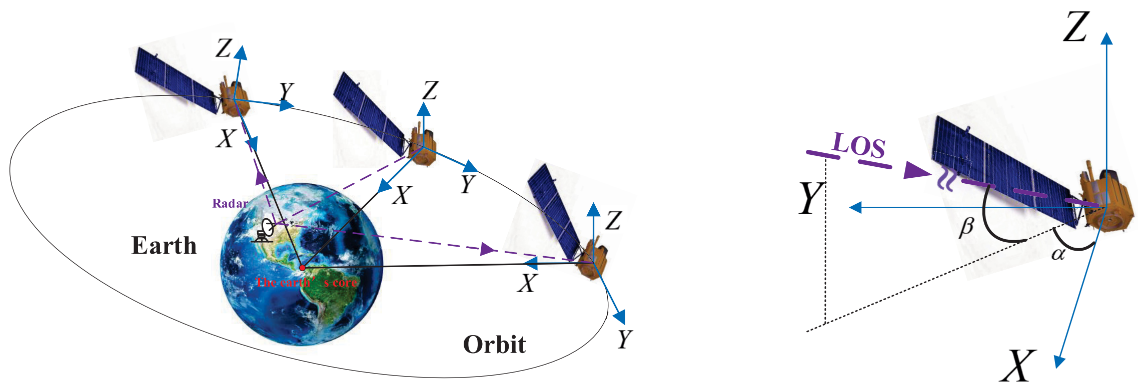
## II. Problem Description and Mathematical Principles

### A. Projection Theory of ISAR Imaging

In this section, we will present the radar observation geometry model of an in-orbit satellite. In the satellite's stabilization coordinate system, the satellite is considered an attitude stabilization target under the attitude controller [23]. As shown in Fig. 1(b), the satellite is assumed to be observed in the local orbital frame $X$–$Y$–$Z$, which is called RTN coordinate system is defined by the radial (R), along-track (T) and cross-track (N) directions [12]. This is a typical coordinate system that provides an essential reference for describing the attitude of an Earth pointing satellite [3], [5]. Under this assumption, the relative rotational motion between the satellite and radar can be simply described by the variation of the radar line of sight (LOS) in the RTN coordinate system. In this way, the ISAR observation geometry is described with the classical turn-table model [16], as shown in Fig. 1(b). In Fig. 1(b), the LOS is defined by the observation angles, i.e., the azimuth angle $\alpha(t_m)$ and the elevation angle $\beta(t_m)$, where $t_m$ is the slow time. The time-varying LOS $\mathbf{i}(t_m)$ can then be represented as

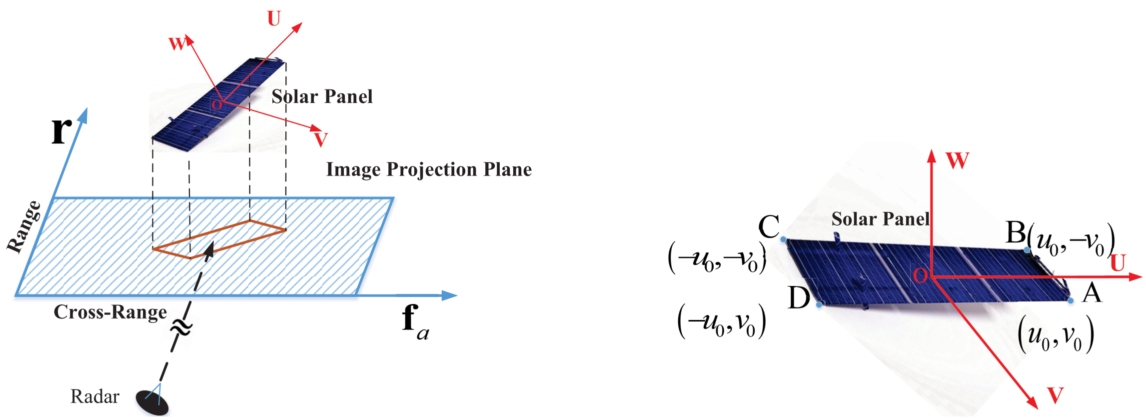
$$\mathbf{i}(t_m) = [\cos\beta(t_m)\sin\alpha(t_m), \cos\beta(t_m)\cos\alpha(t_m), \sin\beta(t_m)]^T. \tag{2}$$

In the satellite's stabilization coordinate system, the observation angles $\alpha(t_m)$ and $\beta(t_m)$ can be obtained from the Kepler two-body motion property [23] and the radar geographic coordinates. Zhou *et al.* [11] provided three methods for solving the radar observation angles, and the second method is referred to in this article. In the turn-table model, the image projection plane, which is defined by the range projection vector $\mathbf{r}$ and the azimuth projection vector $\mathbf{f}_a$ as shown in Fig. 2(a), can be easily determined by the time-varying LOS $\mathbf{i}(t_m)$. According to [11], the projection axis can be expressed as (1), shown at the bottom of the next page, where $\alpha(t_0)$ and $\beta(t_0)$ are the azimuth angle and elevation angle at time $t_0$, respectively, $t_0$ is the observation's starting time, and $\dot{\alpha}$ and $\dot{\beta}$ are first-order




(a) (b)

Fig. 1. (a) Radar observation geometry of satellite in orbit. (b) Satellite-stabilized coordinate.




(a) (b)

Fig. 2. (a) Projection process of ISAR imaging. (b) Solar panel coordinate system.

derivatives of $\alpha(t_m)$ and $\beta(t_m)$, respectively. $\Delta r = (c/2B)$ and $\Delta f_a = (\lambda/2\Omega)$ are the range and cross-range resolutions, where $c$ is the speed of light, $B$ is the bandwidth, $\lambda$ is the wavelength of the emitted signal, and $\mathbf{\Omega}$ is the relative rotation angle between the target and the radar during imaging interval, which can be easily obtained according to the method mentioned in [24].

Assume that the coordinates of an arbitrary point on the solar panel are $\mathbf{P} = [x_p, y_p, z_p]^T$ in the $X$–$Y$–$Z$ coordinate system, and then, the coordinates of its projection on the imaging plane can be expressed as follows:

$$\hat{\mathbf{P}} = \mathbf{MP} \tag{3}$$

where $\hat{\mathbf{P}} = [a_p, b_p]^T$ is the coordinates of the projection point and $\mathbf{M}$ is the projection matrix, which can be expressed as follows:

$$\mathbf{M} = \begin{bmatrix} \mathbf{r}^T \\ \mathbf{f}_a^T \end{bmatrix}. \tag{4}$$

The projection matrix $\mathbf{M}$ can be obtained by the three methods mentioned in [16]. The projection matrix $\mathbf{M}$ links the actual 3-D structure of the satellite target with its 2-D structure in the ISAR image, making it possible to estimate the absolute attitude and size of the satellite from the ISAR image.

### B. Characteristics of the PCA Process for the Solar Panel

PCA is a commonly used data analysis method, which transforms the raw data into a set of orthogonal directions, with which the main components of the data can be extracted.

$$\begin{cases} \mathbf{r} = \dfrac{[\cos\beta(t_0)\sin\alpha(t_0), \cos\beta(t_0)\cos\alpha(t_0), \sin\beta(t_0)]^T}{\Delta r} \\ \mathbf{f}_a = \dfrac{\left[-\sin\beta(t_0)\sin\alpha(t_0)\cdot\dot{\beta} + \cos\beta(t_0)\cos\alpha(t_0)\cdot\dot{\alpha}, -\sin\beta(t_0)\cos\alpha(t_0)\cdot\dot{\beta} - \cos\beta(t_0)\sin\alpha(t_0)\cdot\dot{\alpha}, \cos\beta(t_0)\cdot\dot{\beta}\right]^T}{\Delta f_a} \end{cases} \tag{1}$$

In this section, we will show how to extract the solar panel features using PCA.

Now, we define a new coordinate system $O$-UVW on the solar panel with its origin $O$ at the center of the solar panel, the $U$- and $V$-axes parallel to the long and short edges of the solar panel, respectively. The $W$-axis is perpendicular to the solar panel plane, as shown in Fig. 2(b). Assume that the four vertices of the solar panel are $A$, $B$, $C$, and $D$, located in the four quadrants of the $UOV$ plane, where the coordinates of the vertex $A$ is $[u_0, v_0, 0]^T$. Then, the coordinates of vertexes $B$, $C$, and $D$ are $[u_0, -v_0, 0]^T$, $[-u_0, -v_0, 0]^T$, and $[-u_0, v_0, 0]^T$. Obviously, the solar panel's length and width are $2u_0$ and $2v_0$, respectively. The orientations of long and short edges are $\mathbf{n}_l = [1, 0, 0]^T$ and $\mathbf{n}_s = [0, 1, 0]^T$, respectively. Let $\mathbf{Q}$ be a set of the points on the solar panel, which can be expressed as follows:

$$\mathbf{Q} = \begin{bmatrix} u_1 & u_2 & \cdots & u_N \\ v_1 & v_2 & \cdots & v_N \\ w_1 & w_2 & \cdots & w_N \end{bmatrix} \tag{5}$$

where $N$ is the number of points on the solar panel. In the $O$-UVW coordinate system, since the solar panel can be considered a plane, so $w_i = 0$, $i = 1, 2, \ldots, N$. For a set of points uniformly distributed in a rectangular region, its scattering point coordinates should satisfy a uniform distribution, i.e., as shown in the following equation:

$$\begin{cases} u \sim U(-u_0, u_0) \\ v \sim U(-v_0, v_0). \end{cases} \tag{6}$$

With (6), the mean and variance of the $u$- and $v$-axis coordinates of the point set $\mathbf{Q}$ can be expressed as

$$\begin{cases} \bar{u} = 0, \quad \bar{v} = 0 \\ D_u = \dfrac{u_0^2}{3}, \quad D_v = \dfrac{v_0^2}{3} \end{cases} \tag{7}$$

where $\bar{u}$ and $\bar{v}$ are the mean of $u$ and $v$, respectively, $D_u$ and $D_v$ are the variance of $u$ and $v$, respectively. The PCA process is actually solving the eigenvalues and eigenvectors of the datasets covariance. Here, the covariance of the dataset $\mathbf{Q}$ is expressed as

$$\begin{aligned} \mathbf{C_Q} &= \frac{1}{N}\mathbf{Q}\mathbf{Q}^T \\ &= \frac{1}{N}\begin{bmatrix} \sum_{i=1}^{N} u_i^2 & \sum_{i=1}^{N} u_i v_i & \sum_{i=1}^{N} u_i w_i \\ \sum_{i=1}^{N} u_i v_i & \sum_{i=1}^{N} v_i^2 & \sum_{i=1}^{N} v_i w_i \\ \sum_{i=1}^{N} u_i w_i & \sum_{i=1}^{N} v_i w_i & \sum_{i=1}^{N} w_i^2 \end{bmatrix}. \end{aligned} \tag{8}$$

Since $u$, $v$, and $w$ are independent of each other and $w_i = 0$, (8) can be simplified as

$$\mathbf{C_Q} = \begin{bmatrix} D_u & 0 & 0 \\ 0 & D_v & 0 \\ 0 & 0 & 0 \end{bmatrix} = \mathbf{E}\Lambda\mathbf{E}^T \tag{9}$$

where $\mathbf{E} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is a matrix of eigenvectors and $\Lambda = \begin{bmatrix} D_u & 0 & 0 \\ 0 & D_v & 0 \\ 0 & 0 & 0 \end{bmatrix}$ is a diagonal matrix containing eigenvalues. Notice that the eigenvectors contained by $\mathbf{E}$ are the long and short edges' orientations of the solar panel. At the same time, $\Lambda$ has eigenvalues that correspond to the lengths of the solar panel's long and short edges. In fact, the $O$-UVW coordinate system can be transformed into the $X$–$Y$–$Z$ coordinate system after certain rotational operations. Assume that the coordinates of an arbitrary point in the $O$-UVW coordinate system is $\mathbf{Q}_i = [u_i, v_i, w_i]^T$, and then its coordinates in the $X$–$Y$–$Z$ coordinate system is

$$\mathbf{P}_i = \mathbf{R}(\theta, \varphi, \gamma)\mathbf{Q}_i \tag{10}$$

where $\mathbf{R}(\theta, \varphi, \gamma)$ is the rotation matrix between the $O$-UVW coordinate system and the $X$–$Y$–$Z$ coordinate system [25], [26], $\mathbf{Q}_i$ is an arbitrary point on the solar panel, and $\mathbf{P}_i$ is the corresponding coordinates in the $X$–$Y$–$Z$ coordinate system. The detailed expression for rotation matrix $\mathbf{R}(\theta, \varphi, \gamma)$ is given as follows:

$$\mathbf{R}(\theta, \varphi, \gamma) = \mathbf{R}_z(\theta) \cdot \mathbf{R}_y(\varphi) \cdot \mathbf{R}_x(\gamma) \tag{11}$$

where $\theta$, $\varphi$, and $\gamma$ are the rotational angles around $Z$-, $Y$- and $X$-axes, respectively. $\mathbf{R}_z(\theta)$, $\mathbf{R}_y(\varphi)$, and $\mathbf{R}_x(\gamma)$ can be expressed as follows:

$$\mathbf{R}_z(\theta) = \begin{bmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix} \tag{12}$$

$$\mathbf{R}_y(\varphi) = \begin{bmatrix} \cos\varphi & 0 & \sin\varphi \\ 0 & 1 & 0 \\ -\sin\varphi & 0 & \cos\varphi \end{bmatrix} \tag{13}$$

$$\mathbf{R}_x(\gamma) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\gamma & -\sin\gamma \\ 0 & \sin\gamma & \cos\gamma \end{bmatrix}. \tag{14}$$

Inserting (10) into (3), we have

$$\hat{\mathbf{P}} = \mathbf{M} \cdot \left[\mathbf{R}(\theta, \varphi, \gamma) \cdot \mathbf{Q}\right] = \tilde{\mathbf{M}} \cdot \mathbf{Q} \tag{15}$$

where

$$\tilde{\mathbf{M}} = \mathbf{M} \cdot \mathbf{R}(\theta, \varphi, \gamma) = \begin{bmatrix} \tilde{\mathbf{r}}^T \\ \tilde{\mathbf{f}}_a^T \end{bmatrix} \tag{16}$$

with $\tilde{\mathbf{M}}$ the projection matrix from the $O$-UVW coordinate system to the ISAR imaging plane. Now, we apply the PCA operation on those points in the imaging plane, and the covariance matrix of $\hat{\mathbf{P}}$ can be expressed as

$$\begin{aligned} \mathbf{C}_{\hat{\mathbf{P}}} &= \frac{1}{N}\hat{\mathbf{P}} \cdot \hat{\mathbf{P}}^T = \frac{1}{N}\begin{bmatrix} \sum_{i=1}^{N} a_i^2 & \sum_{i=1}^{N} a_i b_i \\ \sum_{i=1}^{N} a_i b_i & \sum_{i=1}^{N} b_i^2 \end{bmatrix} \\ &= \begin{bmatrix} D_a & D_{ab} \\ D_{ab} & D_b \end{bmatrix} \end{aligned} \tag{17}$$

where $(a_i, b_i)$ is the coordinates of the $i$th scatterer in the ISAR image, $D_a$ and $D_b$ are the variances of range and cross-range coordinates, respectively, and $D_{ab}$ is their covariance.

With covariance matrix $\mathbf{C}_{\hat{\mathbf{P}}}$, we can get its eigenvectors $\mathbf{V}_1$ and $\mathbf{V}_2$ and the eigenvalues $\lambda_1$ and $\lambda_2$. Detailed expressions for $\mathbf{C}_{\hat{\mathbf{P}}}$, $\mathbf{V}_1$, $\mathbf{V}_2$, $\lambda_1$, and $\lambda_2$ are given in Appendix A. Notice (43), where $D_a$, $D_b$, and $D_{ab}$ are functions of the variables $(\theta, \varphi, \gamma, u_0, v_0)$, which makes the eigenvectors and eigenvalues in (45) and (46) also functions of those variables. The variables $(\theta, \varphi, \gamma, u_0, v_0)$ represent the attitude and size of the satellite's solar panel in the $X$–$Y$–$Z$ coordinate system, which is shown in the following equations:

$$[\mathbf{n}_l, \mathbf{n}_s] = \mathbf{R}(\theta, \varphi, \gamma) \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \tag{18}$$

$$L_l = 2u_0, \quad L_s = 2v_0 \tag{19}$$

where $\mathbf{n}_l$ and $\mathbf{n}_s$ are the orientation vectors of long and short edges in $X$–$Y$–$Z$ coordinate system, respectively, and $L_l$ and $L_s$ are the length of the long and short edges, respectively. Equations (18) and (19) provide us a way to obtain the satellite's attitude and size. The detailed procedure will be presented in Section III.

## III. Methodology Description and Parametric Optimization

This section presents specific solar panel attitude and size estimation steps, including the proposed ISAR image segmentation method based on Pix2pixGAN [18] and the PCA-based solar panel features extraction method. In the end, we develop a parametric optimization model to optimally solve the attitude and size parameters by a BFGS-based iteratively searching method.

### A. ISAR Image Segmentation

Since this article focuses on the satellite's solar panels, the first thing we need to do is to separate the solar panel from the ISAR image. Reviewing some traditional segmentation methods, the clustering method can accomplish the segmentation task to some extent. However, the clustering method is based on the texture, color, and brightness of the pixels of different target areas in the image. Since the pixels' characteristics in different parts of the target area are not significantly different in the ISAR images, the clustering method will fail in the ISAR image segmentation task. Nowadays, some deep learning (DL)-based image segmentation methods have excellent performance, such as those based on fully convolutional network (FCN) [27], SegNet [28], and DeepLabV3+ [29]. However, the three methods' loss functions are generally the reconstruction loss function of the whole image, which may lose some pixel-level details. In recent years, Pix2pixGAN has been put forward [18], [30], which has an excellent performance in image-to-image style translation. Compared with the three methods mentioned above, Pix2pixGAN makes the segmented image more accurate in detail by introducing a discriminator with a specific structure called PatchGAN. The discriminator can constrain the generator on small patches rather than on the whole image, making it pay more attention to details.

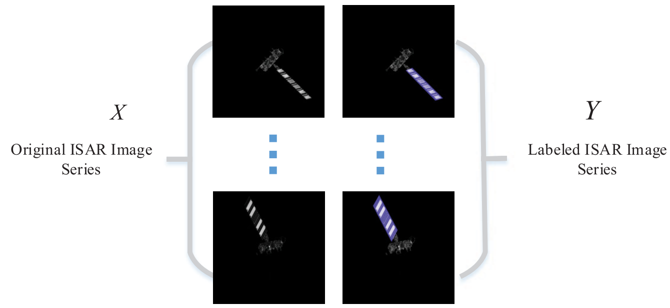



Fig. 3. Original ISAR images of Aqua satellite and the corresponding labeled ISAR images.

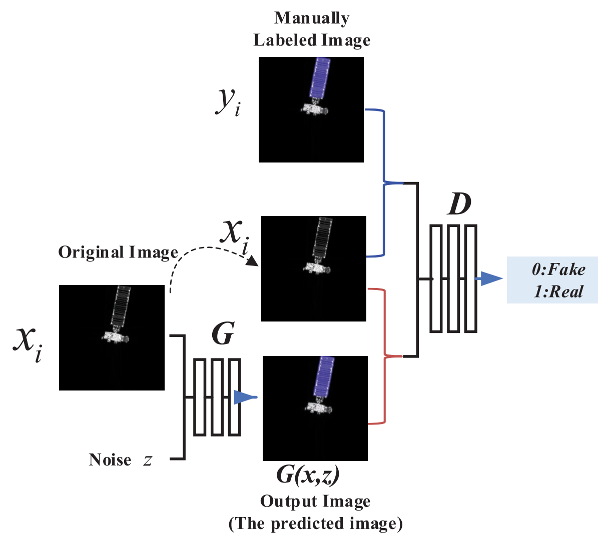



Fig. 4. Training process of Pix2pixGAN.

Inspired by the powerful capability of Pix2pixGAN, the ISAR image segmentation problem can be treated as an image-to-image style translation problem, as shown in Fig. 3. Fig. 3 shows two image sets: the original ISAR images and the labeled ISAR images. Pix2pixGAN can learn a loss function appropriate to train a mapping from the original ISAR image set to the labeled ISAR image set. Then, the output ISAR images, i.e., the images with a labeled solar panel, can be used to separate the solar panel from the initial ISAR images. The original ISAR images and the labeled ISAR images are represented by $X$ and $Y$, respectively. The Pix2pixGAN can automatically learn the mapping function between the two image domains, i.e., $X$ and $Y$. Given training samples $\{x_i\}_{i=1}^N$ and $\{y_i\}_{i=1}^N$, where $x_i \in X$ and $y_i \in Y$, as shown in Fig. 4, the Pix2pixGAN network introduces the idea of generative adversarial, in which the generator $G$ and the discriminator $D$ confront each other during the training process. The generator $G$ generates the predicted labeled images that hard to be distinguished from the real images (the manually labeled images). On the contrary, the discriminator $D$ tries to distinguish the predicted and the manually labeled image. The adversarial process trains the generator to generate realistic-looking (compared to manually labeled images) predictive

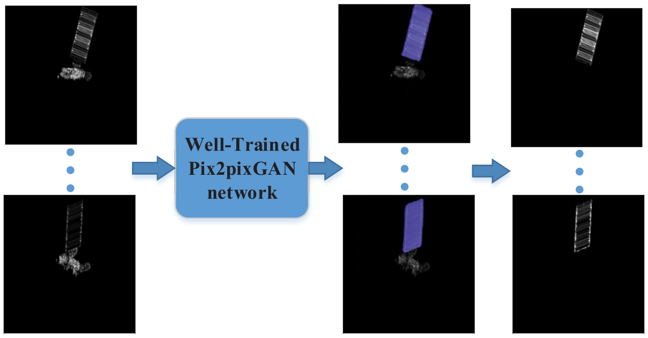



Fig. 5. Flowchart of the proposed ISAR image segmentation method.

images, which can be used to mark the solar panels of satellites in newly incoming ISAR images. Fig. 4 shows the detailed training process. The output image $G(x, z)$, together with the initial image $x$, is one paired image. The labeled image $y$, together with the initial image $x$, is another paired image. These two sets of paired images are inputted into the discriminator to tell whether the generated image $G(x, z)$ can be distinguished from the real labeled image y. With the processing of training, the generator can generate images similar to the labeled images so that the discriminator cannot distinguish them. The loss function of Pix2pixGAN consists of two parts: the traditional adversarial loss function and the $L_1$ loss function, which can be found in [18]. After the training process, the well-trained Pix2pixGAN network can be used to segment new ISAR images of the satellite. The workflow of the proposed segmentation method is shown in Fig. 5.

### B. Point-Set Extraction of the Solar Panel

When we have a segmented image of the solar panel, we need to extract the point set $\tilde{\mathbf{P}}$ from the solar panel using the scattering center extraction method. There are several scattering center extraction methods available, such as the orthogonal matching pursuit (OMP) algorithm [31] and the peak extraction technique [19]. Compared to the peak extraction technique, the OMP algorithm is much more computationally intensive. Considering that the proposed method only utilizes the position information of the point set on the solar panel and to ensure computational efficiency, the peak extraction technique is adopted in this article. In [19], the peak extraction technique combined with the CLEAN technique can be summarized as follows. In each iteration, search the position and amplitude of the point with the highest amplitude, i.e., the peak, in the ISAR image, record the coordinates and amplitude of this point. Remove this point from the original image using the CLEAN technique and then proceed to the next iteration. The process is repeated until the iteration termination condition is reached. After the peak extraction technique, we can get the location matrix $\hat{\mathbf{P}} = \begin{bmatrix} a_1 \cdots a_i \cdots a_N \\ b_1 \cdots b_i \cdots b_N \end{bmatrix}$, which can be used to extract the solar panel's features in Section III-C. The detailed procedure of peak extraction technique is given as follows.

1) *Step 1 (Initialization):* The ISAR image of the satellite's solar panel $\{\mathbf{I}_1(m, n)\}$, where $m$ is the index of range bin, $m = 1, 2, \ldots, M$, and $n$ is the index of cross-range bin, $n = 1, 2, \ldots, N$. The initial location matrix $\hat{\mathbf{P}} = \emptyset$ and the termination conditions (the points number $K$ or the residual energy $\varepsilon$).
2) *Step 2:* Search the peak of $\{\mathbf{I}_i(m, n)\}$ and record its location $(a_i, b_i)$ and amplitude $A_i$, where $\{\mathbf{I}_i(m, n)\}$ is the image of $i$th iteration. Update the location matrix $\hat{\mathbf{P}} = \hat{\mathbf{P}} \cup [a_i, b_i]^T$

$$(a_i, b_i) = \arg\max_{m,n}(\mathbf{I}_i(m, n)) \tag{20}$$

$$A_i = \mathbf{I}_i(a_i, b_i). \tag{21}$$

3) *Step 3:* Remove the extracted peak point $(a_i, b_i)$ in image $\{\mathbf{I}_i(m, n)\}$ by employing the CLEAN technique as follows:

$$\mathbf{I}_{i+1}(m, n) = \mathbf{I}_i(m, n) \odot \mathbf{W}_i(a_i, b_i) \tag{22}$$

where $\odot$ is the Hadamard product, $\mathbf{W}_i(a_i, b_i)$ is the window function of $i$th iteration, which can be expressed as (23), and $\eta_r$ and $\eta_a$ are the radius, usually a few pixels

$$\mathbf{W}_i(a_i, b_i) = \begin{cases} 0, & |m - a_i| < \eta_r, \quad |n - b_i| < \eta_a \\ 1, & \text{others.} \end{cases} \tag{23}$$

4) *Step 4:* Repeat steps 2 and 3 until the number of extracted points becomes $K$ or the amplitude of the extracted point is less than $\varepsilon$.

After the peak extraction process, we can get the point-set location matrix $\hat{\mathbf{P}}$ of the solar panel.

### C. Features Extraction Based on PCA

After obtaining the coordinates of the point set on the solar panel's projection with peak extraction technique, PCA is employed to extract the solar panel's principal components, i.e., the eigenvalues and the eigenvectors of the point-set's covariance matrix, as we have discussed in Section II-B. In general, the first step is to shift each row of the extracted point set matrix $\hat{\mathbf{P}}$ to make it has a zero mean, realized by the following equation:

$$\begin{aligned} \bar{\mathbf{P}} &= \hat{\mathbf{P}} - \text{mean}\left(\hat{\mathbf{P}}\right) \\ &= \begin{bmatrix} \hat{a}_1 - \bar{a} & \hat{a}_2 - \bar{a} & \cdots & \hat{a}_N - \bar{a} \\ \hat{b}_1 - \bar{b} & \hat{b}_2 - \bar{b} & \cdots & \hat{b}_N - \bar{b} \end{bmatrix} \end{aligned} \tag{24}$$

where $\bar{a}$ and $\bar{b}$ denote the mean of $\{a_i\}_{i=1,2,\ldots,N}$ and $\{b_i\}_{i=1,2,\ldots,N}$, respectively. The covariance matrix of $\bar{\mathbf{P}}$ can be expressed as $\mathbf{C}_{\bar{\mathbf{P}}} = (1/N)\bar{\mathbf{P}} \cdot \bar{\mathbf{P}}^T$. Performing an eigenvalue decomposition on $\mathbf{C}_{\bar{\mathbf{P}}}$ yields $\mathbf{C}_{\bar{\mathbf{P}}} = \mathbf{G}\Sigma\mathbf{G}^T$, where $\mathbf{G}$ is the matrix containing eigenvectors and $\Sigma$ is the diagonal matrix containing the eigenvalues. Assume that the extracted eigenvectors and eigenvalues are $\tilde{\mathbf{V}}_1$ and $\tilde{\mathbf{V}}_2$ and $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$, respectively, as shown in Fig. 10. Section III-D will use these extracted features to build a parametric optimization function for estimating the satellite attitude and size.

### D. Parameters Estimation Based on Parametric Optimization

In Section III-C, the extracted eigenvalues and eigenvectors are $\tilde{\lambda}_1$, $\tilde{\lambda}_2$, $\tilde{\mathbf{V}}_1$, and $\tilde{\mathbf{V}}_2$. Assume that their true values are $\lambda_1(\Theta)$, $\lambda_2(\Theta)$, $\mathbf{V}_1(\Theta)$, and $\mathbf{V}_2(\Theta)$, where $\Theta = [\theta, \varphi, \gamma, u_0, v_0]^T$ is the vector of parameters to be estimated, $\theta$, $\varphi$, and $\gamma$ are the rotation angles of the rotation matrix, and $(u_0, v_0)$ are the coordinates of the vertices $A$, all of which we have already described in Section II-B.

Now, we define the angle between the eigenvectors and the $\mathbf{f}_a$-axis as $\phi_i$, which can be represented as

$$\tilde{\phi}_i = \arctan \frac{\tilde{\mathbf{V}}_i(2)}{\tilde{\mathbf{V}}_i(1)} \tag{25}$$

$$\phi_i(\Theta) = \arctan \frac{\mathbf{V}_i(2)}{\mathbf{V}_i(1)} = \arctan \frac{\lambda_i - D_a}{D_{ab}} \tag{26}$$

where $\phi_i$, $\tilde{\phi}_i \in (-(\pi/2), (\pi/2))$, $\tilde{\mathbf{V}}_i(1)$ and $\tilde{\mathbf{V}}_i(2)$ are the first and second elements of eigenvector $\tilde{\mathbf{V}}_i$, and $\mathbf{V}_i(1)$ and $\mathbf{V}_i(2)$ are the first and second elements of eigenvector $\mathbf{V}_i(\Theta)$. We define the difference between $\phi_i(\Theta)$ and $\tilde{\phi}_i$, as well as the difference between $\lambda_i$ and $\tilde{\lambda}_i$ $(i = 1, 2)$, as the optimization function $J(\Theta)$, which is expressed as follows:

$$J(\Theta) = \sum_{f=1}^{F} \left\{ \omega_1 \sum_{i=1}^{2} \left| \phi_i(\Theta) - \tilde{\phi}_i \right| + \omega_2 \sum_{i=1}^{2} \left| \lambda_i(\Theta) - \tilde{\lambda}_i \right| \right\} \tag{27}$$

where $\omega_1$ and $\omega_2$ are the weights, $\omega_1 + \omega_2 = 1$, $\omega_{1,2} \in (0, 1)$, $f$ is the frame index of the ISAR image, $f = 1, 2, \ldots, F$, $F$ the total number of ISAR images. It should be emphasized that in the optimization function $J(\Theta)$, the former term determines the attitude of the satellite's solar panel, whereas the latter term determines the solar panel size, so $\omega_1$ and $\omega_2$ should be well adjusted to optimize the attitude and size estimation. The optimal parameter vector $\hat{\Theta} = [\hat{\theta}, \hat{\varphi}, \hat{\gamma}, \hat{u}_0, \hat{v}_0]^T$ is obtained by minimizing the optimization function $J(\Theta)$, which is represented as follows:

$$\hat{\Theta} = \arg \min_{\Theta} J(\Theta). \tag{28}$$

To solve this optimization, many standard algorithms are available, such as PSO [12], [32] and genetic algorithms. Zhou *et al.* [11] adopted PSO to estimate the solar panel orientation and size in a stepwise way. The problem is that the nonparametric methods search for optimal solutions in an exhaustive way, which is computationally expensive, and not very accurate. In this article, to avoid the accumulation of errors due to the stepwise estimation and to consider the computational complexity and accuracy, we adopt an iterative BFGS-based algorithm with a quasi-Newton iteration.

First, we need to calculate the partial derivatives of each parameter in the parameter vector $\Theta$. The detailed procedure for gradient derivation is in Appendix B. With the partial derivative expressions (47)–(70), the gradient of loss function $J(\Theta)$ for $\Theta$ is

$$\nabla \mathbf{J} = \left[ \frac{\partial J(\Theta)}{\partial \theta}, \frac{\partial J(\Theta)}{\partial \varphi}, \frac{\partial J(\Theta)}{\partial \gamma}, \frac{\partial J(\Theta)}{\partial u_0}, \frac{\partial J(\Theta)}{\partial v_0} \right]^T. \tag{29}$$

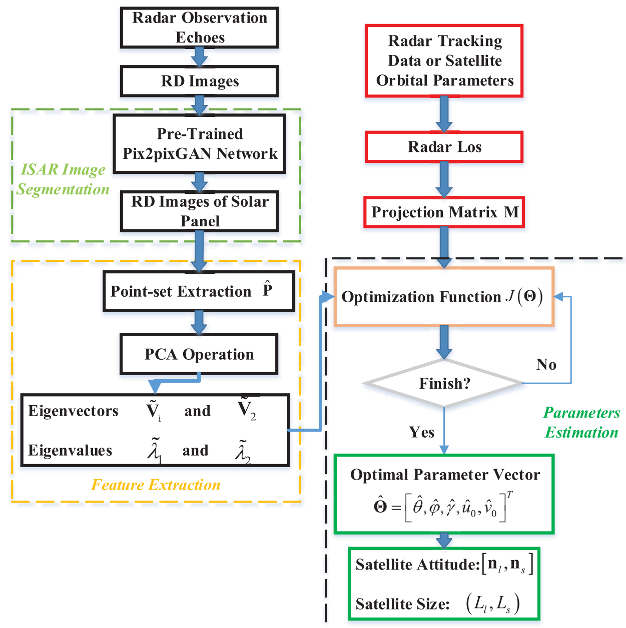



Fig. 6. Flowchart of the proposed method.

In general, the second-order derivative is needed to build the Hessian matrix if gradient descent algorithms are applied here. Considering the computation complexity of second-order derivatives, the BFGS algorithm is adopted in this article to search for the optimal solution. Let $\Theta_0 = [0, 0, 0, 0, 0]^T$ be the initial parameter vector and $\Theta_k$ be the parameter vector of the $kth$ iteration. The searching direction in BFGS is updated as follows:

$$\mathbf{d}_k = -\mathbf{B}_k \nabla \mathbf{J}_k. \tag{30}$$

Then, the $k + 1th$ parameter vector can be expressed as

$$\Theta_{k+1} = \Theta_k + \lambda_k \mathbf{d}_k \tag{31}$$

where $\lambda_k$ is the searching step that can be estimated by (32) via some 1-D inexact searching methods, such as golden-section search or the Armijo–Goldstein stepsize rule

$$\lambda_k = \arg \min_{\lambda_k} (J(\Theta_k + \lambda_k \mathbf{d}_k)). \tag{32}$$

The Hessian matrix $\mathbf{B}_k$ in BFGS is updated as follows:

$$\mathbf{B}_{k+1} = \mathbf{B}_k + \frac{\mathbf{y}_k \cdot \mathbf{y}_k^T}{\mathbf{y}_k \cdot \mathbf{s}_k^T} - \frac{\mathbf{B}_k \mathbf{s}_k \mathbf{s}_k^T \mathbf{B}_k}{\mathbf{s}_k^T \mathbf{B}_k \mathbf{s}_k} \tag{33}$$

where $\mathbf{s}_k = \lambda_k \mathbf{d}_k$ and $\mathbf{y}_k = \nabla \mathbf{J}_{k+1} - \nabla \mathbf{J}_k$. The iteration can be terminated when $\|\nabla \mathbf{J}_k\| < \varepsilon$ or $|J(\Theta_{k+1}) - J(\Theta_k)| < \varepsilon$.

When the iteration terminates, the optimal parameter vector $\hat{\Theta} = [\hat{\theta}, \hat{\varphi}, \hat{\gamma}, \hat{u}_0, \hat{v}_0]^T$ is determined. Next, the orientations of long and short edges of the solar panel as well as their sizes are obtained by some simple calculations, shown as (18) and (19).

### E. Flowchart of the Proposed Method

Fig. 6 shows the flowchart of the proposed method. From the figure, it can be seen that the proposed algorithm is

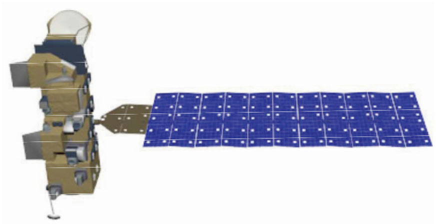


Fig. 7. CAD model of Aqua satellite.

TABLE I
RADAR PARAMETERS AND LOCATION

| | | |
|---|---|---|
| Radar Location | (116.4E,39.9N,0m) | |
| Radar Parameters | Center Frequency | 16.67GHz |
| | Pulse Repetition Frequency | 25Hz |
| | Sample Frequency | 2.4GHz |
| | Bandwidth | 2GHz |
| Image Resolution | 0.075*0.075 | |

divided into three main parts. The first step is the ISAR image segmentation, where the satellite ISAR images are segmented according to the well-trained Pix2pixGAN network to obtain the solar panel images. The second step is the PCA-based feature extraction of the solar panel for obtaining the feature values and eigenvectors of the solar panel point sets. Finally, based on the extracted features and the ISAR imaging projection matrix, the proposed algorithm builds an optimization function, and the attitude and size parameters are solved with a fast iterative gradient method.

## IV. EXPERIMENTAL ANALYSIS

### A. Introduction of Experimental Data

Before we begin the experiment, a brief description of the experimental data is provided. Due to the lack of measured satellite data, the satellite radar echoes used in the experiment are EM simulation data, generated by adopting the physical optics (PO) algorithm proposed in [33]. The computer-aided-design (CAD) model of the Aqua satellite is given in Fig. 7, which is available at nasa3d.arc.nasa.gov/models. A Ku-band radar is adopted to generate the radar echoes in this article, whose parameters are shown in Table I. The azimuth angle $\alpha(t_m)$ and elevation angle $\beta(t_m)$ of the radar LOS is generated by Satellite Tool Kit (STK), which is consistent with the approach taken in [11] and [17]. The orbits of Aqua satellite are shown in Fig. 8. The purple lines are the segment of the trajectory that can be observed by radar. It should be emphasized that, in practice, the radar LOS can be derived either from radar tracking data or from satellite orbital parameters, referring to the three methods mentioned in [17] and [37]. The satellite is considered a three-axis stabilized target in the experiment, which is stably placed in the $X$–$Y$–$Z$

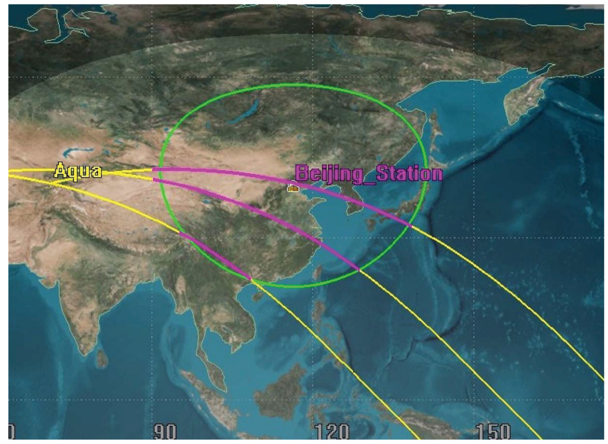



Fig. 8. Orbits of Aqua satellite simulated by STK.

TABLE II
DURATION OF RADAR OBSERVATIONS AND NUMBER OF ISAR IMAGES ACQUIRED

| | Observation Time [s] | Image Size | Image Number |
|---|---|---|---|
| Lap1 | 708.22 | | 51 |
| Lap2 | 679.78 | $512 \times 512$ | 49 |
| Lap3 | 473.24 | | 42 |

coordinate system. STK is used to generate the LOS of the satellite continuously, which lasts for three laps. Based on the LOS information, radar echoes of the three laps observation are generated according to the PO algorithm [33]. Based on the acquired EM simulation echoes, the ISAR images can be obtained by using the conventional range-Doppler (RD) algorithm [35]. After acquiring three laps of satellite ISAR images, the first two laps of ISAR images will be used as training data to train a good ISAR image segmentation network Pix2pixGAN. In Table II, we give the observation durations of the three laps of data, as well as the number of images acquired per lap and the image size. It should be emphasized that more data can be used for training the network in the practical case to ensure the good performance of the network. The third lap of ISAR images is used as test data to estimate the satellite's attitude and size.

### B. Performance of the Proposed Method

In this section, the performance of the proposed method will be evaluated with simulated data of Aqua satellite. As we have mentioned in Section III, the first step is to separate the solar panel from the original ISAR image. Fig. 9(a)–(c) shows the original ISAR images of the Aqua satellite and Fig. 9(d)–(f) shows the outputs of the Pix2pixGAN network. As one can note, the solar panel area in the image is perfectly colored with blue. By selecting the colored pixels of images in Fig. 9(d)–(f),

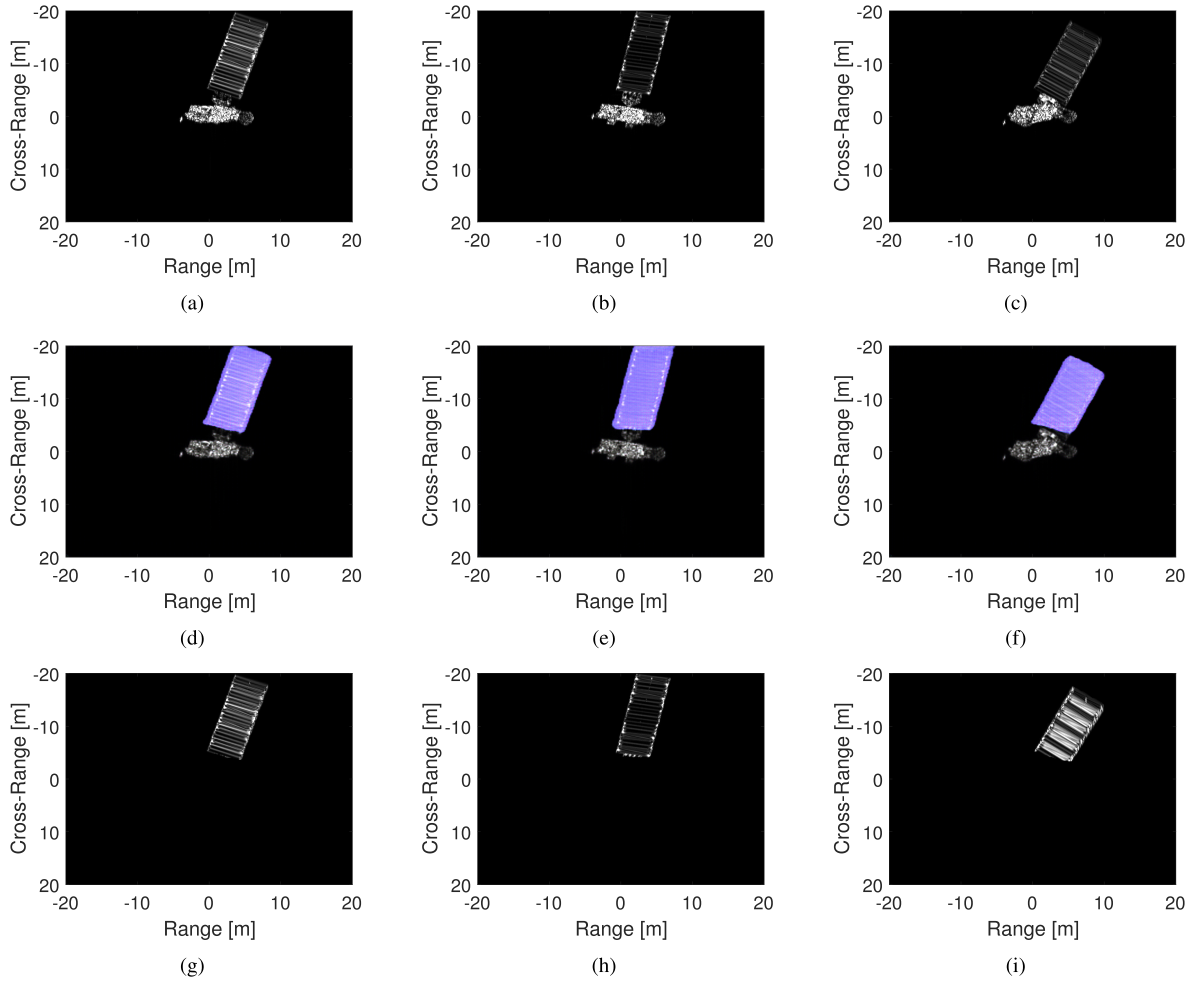



Fig. 9. Some examples of the segmentation outputs for poses I–III. (a)–(c) Examples of original ISAR images for poses I–III, respectively. (d)–(f) Corresponding outputs of the Pix2pixGAN network (the segmented ISAR images). (g)–(i) Corresponding solar panel parts of Aqua satellite.

we can get the solar panel part of the Aqua satellite as shown in Fig. 9(g)–(i).

After acquiring the satellite's solar panel image, the points on the satellite's solar panel are extracted using the peak extraction technique to obtain the point set belonging to the satellite solar panel, as shown in Fig. 10(a)–(c). Based on the acquired locations of the point set, we can obtain the location matrix $\hat{\mathbf{P}}$. The location matrix $\hat{\mathbf{P}}$ will undergo a PCA operation, which has been mentioned in Section III-C. Then, the eigenvalues $\tilde{\lambda}_1$ and $\tilde{\lambda}_2$, as well as the eigenvectors $\tilde{\mathbf{V}}_1$ and $\tilde{\mathbf{V}}_2$ of the solar panel, are obtained. The extracted eigenvectors and eigenvalues are shown in Fig. 10(d)–(f).

After the feature extraction process, we can get a series of eigenvectors $\tilde{\mathbf{V}}^i_{1,2}$ and eigenvalues $\tilde{\lambda}^i_{1,2}$, where $i$ is the image index, $i = 1, 2, \ldots, F$, $F$ the total number of ISAR images. These eigenvectors and eigenvalues are input into the optimization function $J(\Theta)$ to perform a fast search for the satellite's attitude and size parameters by the quasi-Newtonian iterative search method proposed in Section III-D. For comparison, we also provide the search results by the PSO algorithm. In the searching process, the termination condition is $|J(\Theta_{k+1}) - J(\Theta_k)| < \varepsilon$ for both PSO and the proposed method, where $\varepsilon = 10^{-4}$. The variation curves of the optimization function value with the number of iterations are shown in Fig. 11(a), and the variation curves of the attitude and size errors compared with true values are shown in Fig. 11(b) and (c). The optimal estimation results for the satellite's attitude and size are listed in Table III. From Fig. 11(a), we can see that the proposed gradient iteration-based fast search algorithm converges faster and requires fewer iterations to reach convergence than the PSO search algorithm. During the estimation, the iteration number is set to be 400 for both algorithms, which is fully sufficient for the fitness function to converge to the optimal solution. As we can see from Fig. 11(a)–(c), the proposed method requires less than 40 iterations to reach convergence, whereas the PSO search method requires more than 150 iterations. From the optimal estimation results given in Table III, the estimation results of the proposed method are more accurate. Attitude errors for both the long and short edges estimated by the proposed method are around 1°, while those estimated by the PSO-based algorithm are more than 5°. The size errors for both long and short edges estimated by the proposed method are less than 1 m, while those estimated by the PSO-based algorithm are somewhat larger.

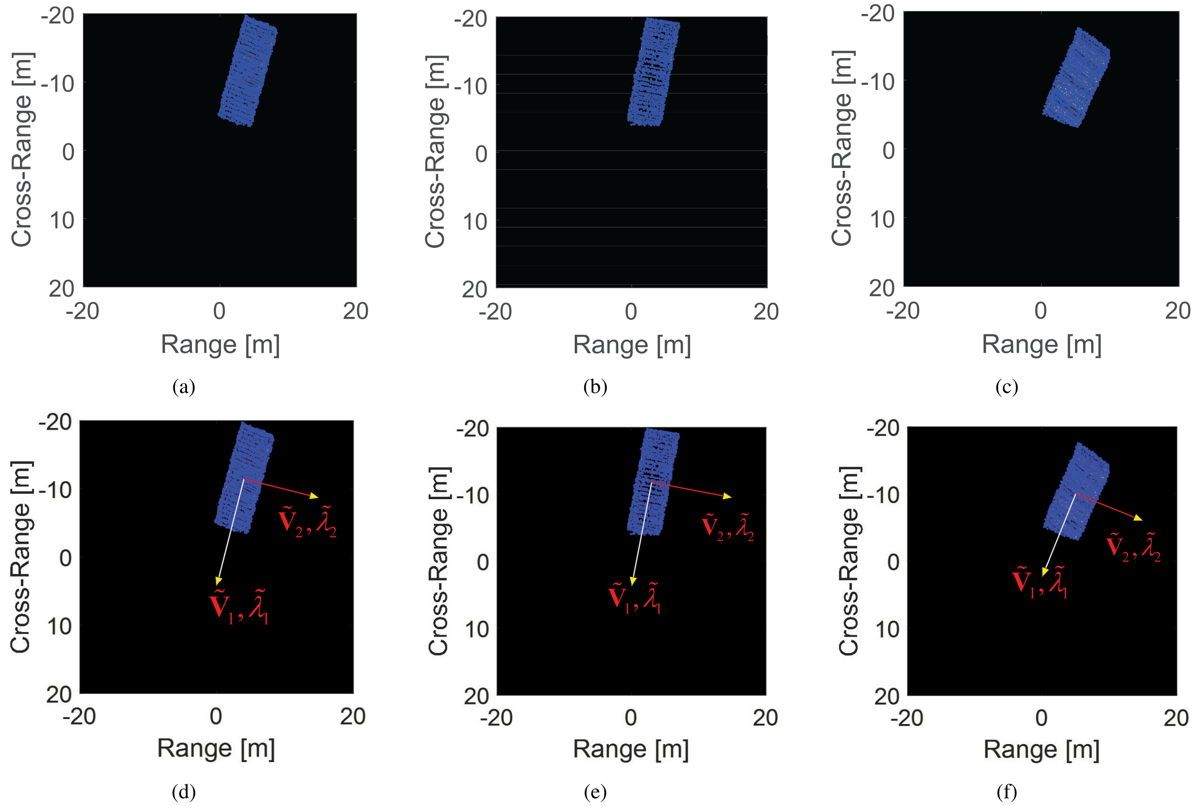



Fig. 10. Feature extraction of satellite's solar panel (poses I–III). (a)–(c) Solar panel part of satellite and the corresponding point set. (d)–(f) Extracted eigenvalues and the eigenvectors.

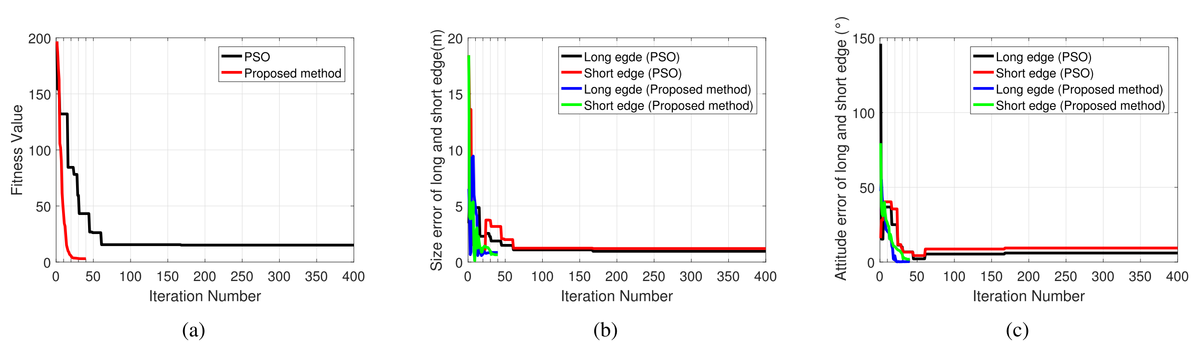



Fig. 11. Cost function value, attitude, and size error variation curves during iteration with the proposed method and PSO algorithm for pose I. (a) Cost function values with the number of iterations. (b) Size errors with the number of iterations for the long and short edges (the red and black lines are the results of PSO algorithm, and the blue and green lines are the results of the proposed method). (c) Attitude errors with the number of iterations for the long and short edges.

### *C. Comparison Analysis*

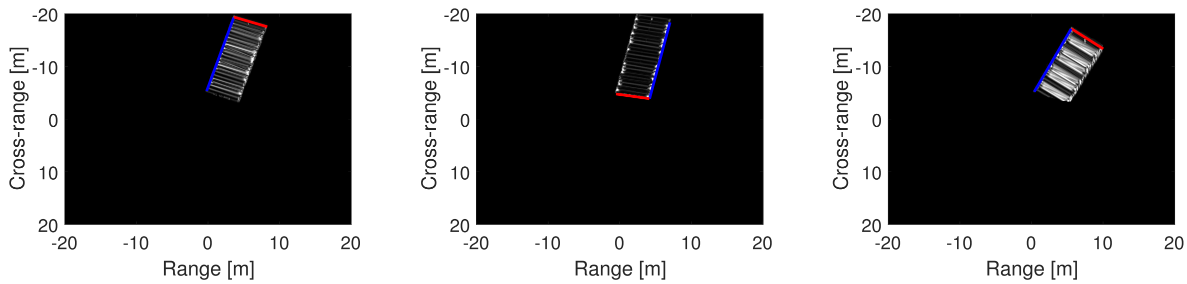
In this section, to highlight the advantages and effectiveness of the proposed method, the proposed method's performance is compared to the algorithm proposed in [11] and the traditional factorization-based reconstruction method [14]

$$\begin{cases} \mathbf{k}(\alpha_L, \beta_L) = [\cos\beta_L \sin\alpha_L, \cos\beta_L \cos\alpha_L, \sin\beta_L]^T \\ \mathbf{k}(\alpha_S, \beta_S) = [\cos\beta_S \sin\alpha_S, \cos\beta_S \cos\alpha_S, \sin\beta_S]^T. \end{cases} \quad (35)$$

First of all, the performance of the algorithm proposed in [11] is investigated. Zhou *et al.* [11] taken advantage of the line features, i.e., the solar panels' long and short edges. Zhou *et al.* [11] extracted the line features of long and short edges by the Randon transform and association method and then searched the attitude of long and short edges by PSO. The optimization function is (34), as shown at the bottom of the next page, where $(\alpha_L, \alpha_S, \beta_L, \beta_S)$ are the attitude parameters to be evaluated, $\alpha_L$ and $\beta_L$ are the azimuth angle and elevation of the long edge, $\alpha_S$ and $\beta_S$ are the azimuth angle and elevation angle of the short edge, and $\mathbf{k}(\alpha_L, \beta_L)$ and $\mathbf{k}(\alpha_S, \beta_S)$ are the attitude vectors of long and short edges, as shown in (35). $\mathbf{M}$ is the projection matrix, $\hat{\psi}_{Lf}$ and $\hat{\psi}_{Sf}$ are the angles of the long and short edges to the Doppler axis, respectively, and $a_1 = 0.6$ and $a_2 = 0.95$ are the balance factors. Based on the estimated attitude parameters of long and short edges, their sizes can also be estimated by using the PSO algorithm.

TABLE III
ESTIMATED ATTITUDES AND SIZES OF AQUA SATELLITE BY DIFFERENT METHODS AT POSE I

| **Pose I** | | True values | Proposed method | | Method in [11] | Factorization -based method |
|---|---|---|---|---|---|---|
| | | | BFGS | PSO | | |
| Attitude | Long edge | (-0.11,0.98,-0.18) | (-0.11,0.98,-0.18) | (-0.03,0.99,-0.11) | (-0.21,0.97,-0.12) | Unknown |
| | Error(degree) | \ | 0.28 | 6.03 | 6.39 | Unknown |
| | Short edge | (-0.90,-0.18,-0.39) | (-0.92,-0.17,-0.36) | (-0.96,-0.06,-0.28) | (-0.83,-0.23,-0.51) | Unknown |
| | Error(degree) | \ | 1.87 | 9.38 | 8.55 | Unknown |
| Size | Long edge | 15.00 | 15.85 | 15.96 | 15.97 | 12.93 |
| | Error(meter) | \ | 0.85 | 0.96 | 0.97 | 2.07 |
| | Short edge | 6.16 | 6.81 | 7.36 | 5.82 | 5.14 |
| | Error(meter) | \ | 0.65 | 1.20 | 0.34 | 1.02 |




(a) (b) (c)

Fig. 12. Extraction and association of long (blue lines) and short (red lines) edges among the ISAR images sequence: (a) pose I, (b) pose II, and (c) pose III.

The optimization function is (36), where $l_L$ and $l_S$ are the length parameters to be estimated and $\text{length}_{Lf}$ and $\text{length}_{Sf}$ are the lengths of long and short edges in ISAR images. For convenience, in this article, we extract the line features of the long and short edges in segmented solar panel images as shown in Fig. 12, where the blue lines are the long edges and red lines are the short edges. The estimated attitude and size parameters are listed in Table III. As can be seen from the table, compared to the proposed method in this article, the estimated attitude error for both long and short edges is more than 5° by the approach in [11], and the size estimation error of the long edge is also slightly worse than that of the proposed method. This is because the proposed method treats the solar panel as a rectangular whole, and unlike the approach in [11], the proposed algorithm can find the optimal value more accurately by using a fast gradient-based search method

$$\begin{cases} \min\limits_{l_L} \sum\limits_{i=1}^{F} \left| \|l_L \mathbf{M}_i \cdot \mathbf{k}(\alpha_L, \beta_L)\|_2 - \text{length}_{Li} \right| \\ \min\limits_{l_S} \sum\limits_{i=1}^{F} \left| \|l_S \mathbf{M}_i \cdot \mathbf{k}(\alpha_S, \beta_S)\|_2 - \text{length}_{Si} \right|. \end{cases} \tag{36}$$

Next, the traditional factorization method is adopted to reconstruct the 3-D construction of the Aqua satellite, and the size of the satellite's solar panel can be estimated by measuring the distance between the solar panel vertices. The factorization method's basic idea is to reconstruct target's 3-D scattering point structure by decomposing the trajectory matrix $\mathbf{T}$ via the singular value decomposition (SVD) algorithm. The trajectory matrix $\mathbf{T}$ is obtained by associating scatterers from ISAR images sequence, which can be expressed as

$$\mathbf{T} = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1Q} \\ \vdots & \vdots & \ddots & \vdots \\ a_{F,1} & a_{F,2} & \cdots & a_{F,Q} \\ b_{1,1} & b_{1,2} & \cdots & b_{1,Q} \\ \vdots & \vdots & \ddots & \vdots \\ b_{F,1} & b_{F,1} & \cdots & b_{F,Q} \end{bmatrix} \tag{37}$$

where $(a_{i,j}, b_{i,j})$ is the $j$th scatterer's position in the $i$th frame of ISAR images, $i = 1, 2, \ldots, F$, $F$ the total number of ISAR images, and $j = 1, 2, \ldots, Q$, $Q$ the total number of scatterers.

$$\min_{(\alpha_L,\alpha_S,\beta_L,\beta_S)} \sum_{i=1}^{F} \left( \left\| \mathbf{M}_i \cdot \mathbf{k}(\alpha_L, \beta_L) - \left(\sin \hat{\psi}_{Li}, \cos \hat{\psi}_{Li}\right)^T \right\|_2 + a_1 \left\| \mathbf{M}_i \cdot \mathbf{k}(\alpha_S, \beta_S) - \left(\sin \hat{\psi}_{Si}, \cos \hat{\psi}_{Si}\right)^T \right\|_2 \right) + a_2\{\tan \alpha_L \tan \alpha_S + \cos(\beta_L - \beta_S)\} \tag{34}$$

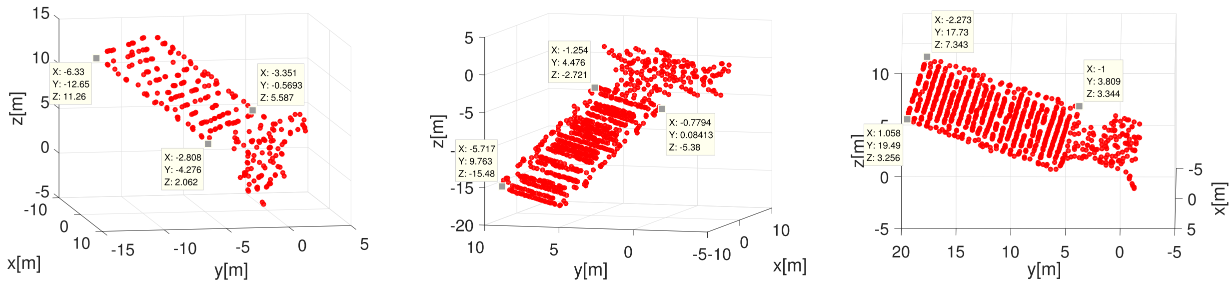



Fig. 13. Reconstruction results of Aqua satellite. (a) Reconstruction result of pose I. (b) Reconstruction result of pose II. (c) Reconstruction result of pose III.

By decomposing $\mathbf{T}$ via SVD, we can get

$$\mathbf{T} = \mathbf{U}\Sigma\mathbf{V} = \left(\mathbf{U}\sqrt{\Sigma}\right)\left(\sqrt{\Sigma}\mathbf{V}\right) = \mathbf{HP}. \tag{38}$$

The decomposition of (38) is not unique, and for any $\mathbf{BB}^{-1} = \mathbf{I}$, $\mathbf{HP} = (\mathbf{HB})(\mathbf{B}^{-1}\mathbf{P})$. Under the constraints of (39), $\hat{\mathbf{H}} = \mathbf{HB}$, and $\hat{\mathbf{P}} = \mathbf{B}^{-1}\mathbf{P}$ can be one acceptable solution, where $\hat{\mathbf{H}} = [\mathbf{f}_{a1}, \mathbf{f}_{a2}, \ldots, \mathbf{f}_{aF}, \mathbf{r}_1, \mathbf{r}_2, \ldots, \mathbf{r}_F]^T$ and $\hat{\mathbf{P}}$ is the position vectors of 3-D scatterers of the target. After getting $\hat{\mathbf{P}}$, the size of the satellite's solar panel can be obtained with some manual effort by measuring the distance between the vertices. Fig. 13 shows the reconstructed results of Aqua, from which we can see the target's rough structure and shape. The estimation results are listed in Table III. It can be seen from the results that, compared to the proposed algorithm and the method proposed in [11], the factorization-based approach gives the worst estimation results and the attitude of the satellite cannot be obtained due to the presence of an unknown rotation matrix $\mathbf{B}$

$$\begin{cases} \|\mathbf{r}_k\| = 1 \\ \|\mathbf{f}_{ak}\| = 1 \\ \mathbf{r}_k \cdot \mathbf{f}_{ak} = 0. \end{cases} \tag{39}$$

The above comparison between the proposed algorithm and the existing algorithm shows the proposed algorithm's effectiveness and accuracy. To analyze the effect of the noise on the proposed algorithm and the comparison algorithm, different Gaussian white noise is added to the ISAR images, and the signal-to-noise ratio (SNR) is varied from −20 to 20 dB to analyze the performance of the proposed method and the method proposed in [11]. It should be noted that the noise also affects the ISAR image segmentation introduced in Section.III-A. Since the focus of the analysis here is on the effect of noise on the parameter estimation accuracy, the impact of noise on image segmentation is not discussed. In the case of a serious impact of noise on image segmentation, manual assistance for segmentation can be considered. Also, due to the traditional factorization method's poor performance, no further comparison is made here. Fig. 14 shows the error variation of the proposed method and the method in [11] for estimating the satellite's attitude and size under different SNRs. From Fig. 14, we can see that both of the methods perform well under low SNRs. For the method proposed in [11], the algorithm has good accuracy in estimating the attitude and size when the SNR is not less than −5 dB.

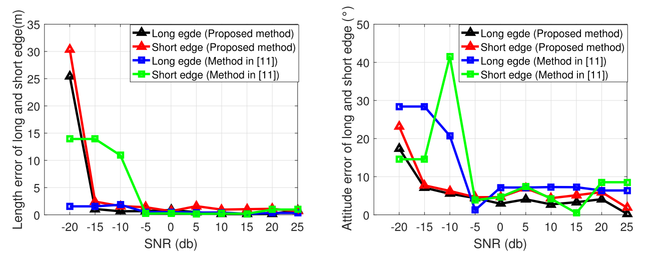



Fig. 14. Variation curves of estimation errors under different SNR conditions for pose I. (a) Estimation error of the satellite's attitude. (b) Estimation error of the satellite's size.

For the proposed method, the estimation accuracy stays high when SNR is not less than −15 dB. We find that since the proposed method utilizes the point features during the experiments, it is more robust under low SNR conditions than the line features used in [11]. When the SNR is getting lower, the line features will be difficult to be extracted. It can be seen that the proposed algorithm has good antinoise interference capability.

From the comparison of the proposed method with existing typical methods, it can be seen that the proposed method performs better and is able to estimate satellite attitude and size simultaneously with higher estimation accuracy than other methods.

### D. Performance Investigation

In general, the proposed algorithm performs well in most of the cases. Combined with the solar panel's rectangular structure, the proposed algorithm can perform joint fast estimation of satellite's attitude and size with high estimation accuracy and noise robustness, verified by the previous experimental analysis. To further demonstrate the effectiveness of the proposed algorithm, we provide the satellite attitude and size estimation results at two other poses, as well as the estimation results of the other two compared methods, which are listed in Tables IV and V, respectively. From the table, it can be seen that the proposed method and the method in [11] perform well when the satellite is at different poses. Besides, the proposed method has a slight advantage in the accuracy of the estimation parameters. However, the estimation results of the factorization-based method are less

TABLE IV
ESTIMATED ATTITUDES AND SIZES OF AQUA SATELLITE BY DIFFERENT METHODS AT POSE II

| **Pose II** | | True values | Proposed method | | Method in [11] | Factorization -based method |
|---|---|---|---|---|---|---|
| | | | BFGS | PSO | | |
| Attitude | Long edge | (0.61,-0.77,-0.17) | (0.61,-0.77,-0.18) | (0.60,-0.78,-0.18) | (0.62,-0.77,-0.16) | Unknown |
| | Error(degree) | \ | 1.03 | 0.82 | 0.51 | Unknown |
| | Short edge | (0.79,0.56,0.22) | (0.79,0.57,0.23) | (0.79,0.62,-0.01) | (0.79,0.59,0.18) | Unknown |
| | Error(degree) | \ | 0.65 | 13.60 | 2.60 | Unknown |
| Size | Long edge | 15.00 | 15.16 | 15.21 | 14.82 | 14.52 |
| | Error(meter) | \ | 0.16 | 0.21 | 0.18 | 0.48 |
| | Short edge | 6.16 | 6.69 | 5.81 | 5.47 | 5.16 |
| | Error(meter) | \ | 0.53 | 0.35 | 0.69 | 1.00 |

TABLE V
ESTIMATED ATTITUDES AND SIZES OF AQUA SATELLITE BY DIFFERENT METHODS AT POSE III

| **Pose III** | | True values | Proposed method | | Method in [11] | Factorization -based method |
|---|---|---|---|---|---|---|
| | | | BFGS | PSO | | |
| Attitude | Long edge | (-0.67,0.62,0.42) | (-0.67,0.61,0.43) | (-0.69,0.61,0.38) | (-0.63,0.61,0.48) | Unknown |
| | Error(degree) | \ | 0.71 | 2.24 | 4.15 | Unknown |
| | Short edge | (-0.64,-0.76,0.09) | (-0.66,-0.75,0.06) | (-0.67,-0.74,0.03) | (-0.65,-0.76,0.06) | Unknown |
| | Error(degree) | \ | 2.12 | 6.86 | 2.02 | Unknown |
| Size | Long edge | 15.00 | 15.38 | 15.29 | 15.64 | 14.54 |
| | Error(meter) | \ | 0.38 | 0.29 | 0.64 | 0.46 |
| | Short edge | 6.16 | 6.78 | 7.02 | 6.83 | 5.56 |
| | Error(meter) | \ | 0.62 | 0.96 | 0.67 | 0.60 |

satisfactory, the reconstruction results are shown in Fig. 13. The experiments with different poses fully illustrate the excellent performance of the proposed algorithm.

However, we also found some critical factors that affect the proposed algorithm's performance during the experiments.

First, the proposed algorithm depends on the integrity of the solar panel. In some extreme cases, inadequate imaging algorithms or mutual occlusion between targets can lead to incomplete ISAR images of the satellite, which in turn can lead to incomplete solar panels and can eventually seriously affect the performance of the proposed algorithm.

Second, the proposed algorithm also suffers from the factor mentioned in [11], that is, under certain viewing angles, the solar panel will be projected as a narrow quadrilateral or even a line. In this case, the proposed method will not be able to extract enough point features, failing of the algorithm. The above two problems can be solved by obtaining more data from long-time observations or by setting up a multistation radar observation to eliminate the data that are obscured or projected as a narrow quadrilateral and improve the proposed algorithm's performance.

## V. CONCLUSION

In this article, an innovative method for the joint estimation of a satellite's solar panel attitude and size is presented. With prior LOS parameters, a projection matrix during ISAR imaging of the satellite is created to associate the satellite's absolute attitude and absolute size with the extracted features in the ISAR images. Taking advantage of the satellite solar panel's rectangular structure, we analyzed the relationship between the eigenvectors and the eigenvalues of the point set on the solar panel and its attitude and size by PCA and established an optimization function for estimating satellite attitude and size parameters. By deriving the optimization function gradient, a fast iterative search algorithm based on BFGS is adopted to perform rapid joint estimation of satellite attitude and size. The experimental results show the effectiveness of the proposed algorithm.

## Appendix A

Considering (2) and (4), the expressions of $\tilde{\mathbf{r}}^T$ and $\tilde{\mathbf{f}}_a^T$ are given as follows:

$$\begin{cases} \tilde{\mathbf{r}}^T = \mathbf{r}^T \cdot \mathbf{R}(\theta, \varphi, \gamma) \\ \tilde{\mathbf{f}}_a^T = \mathbf{f}_a^T \cdot \mathbf{R}(\theta, \varphi, \gamma). \end{cases} \tag{40}$$

Assume that $\tilde{\mathbf{r}}^T$ and $\tilde{\mathbf{f}}_a^T$ can be expressed as

$$\begin{cases} \tilde{\mathbf{r}}^T = [\tilde{r}_1, \tilde{r}_2, \tilde{r}_3] \\ \tilde{\mathbf{f}}_a^T = \left[\tilde{f}_1, \tilde{f}_2, \tilde{f}_3\right]. \end{cases} \tag{41}$$

Then, according to (15), we have

$$\begin{cases} a_i = \tilde{r}_1 \cdot u_i + \tilde{r}_2 \cdot v_i + \tilde{r}_3 \cdot w_i \\ b_i = \tilde{f}_1 \cdot u_i + \tilde{f}_2 \cdot v_i + \tilde{f}_3 \cdot w_i. \end{cases} \tag{42}$$

Considering $w_i = 0$ in the $O$-UVW coordinate system, thus the variances of $a$ and $b$, as well as their covariance, can be expressed as

$$\begin{cases} D_a = \tilde{r}_1^2 \cdot D_u + \tilde{r}_2^2 \cdot D_v \\ D_b = \tilde{f}_1^2 \cdot D_u + \tilde{f}_2^2 \cdot D_v \\ D_{ab} = \tilde{r}_1 \tilde{f}_1 \cdot D_u + \tilde{r}_2 \tilde{f}_2 \cdot D_v. \end{cases} \tag{43}$$

Bringing (43) into (17) yields

$$\mathbf{C}_{\hat{\mathbf{P}}} = \begin{bmatrix} D_a & D_{ab} \\ D_{ab} & D_b \end{bmatrix}. \tag{44}$$

The eigenvalue decomposition of covariance matrix $\mathbf{C}_{\hat{\mathbf{P}}}$ yields eigenvalues $\lambda_1$ and $\lambda_2$ and eigenvectors $\mathbf{V}_1$ and $\mathbf{V}_2$, which are

$$\begin{cases} \lambda_1 = \dfrac{(D_a + D_b) + \sqrt{(D_a + D_b)^2 - 4\left(D_a D_b - D_{ab}^2\right)}}{2} \\ \lambda_2 = \dfrac{(D_a + D_b) - \sqrt{(D_a + D_b)^2 - 4\left(D_a D_b - D_{ab}^2\right)}}{2} \end{cases} \tag{45}$$

$$\mathbf{V}_1 = \begin{bmatrix} \dfrac{1}{\sqrt{1 + \left(\frac{\lambda_1 - D_a}{D_{ab}}\right)^2}} \\ \dfrac{1}{\sqrt{1 + \left(\frac{D_{ab}}{\lambda_1 - D_a}\right)^2}} \end{bmatrix}, \quad \mathbf{V}_2 = \begin{bmatrix} \dfrac{1}{\sqrt{1 + \left(\frac{\lambda_2 - D_a}{D_{ab}}\right)^2}} \\ \dfrac{1}{\sqrt{1 + \left(\frac{D_{ab}}{\lambda_2 - D_a}\right)^2}} \end{bmatrix}. \tag{46}$$

## Appendix B

Considering that the eigenvalues in (45) and the eigenvectors in (46) are both composed of $D_a$, $D_b$, and $D_{ab}$, the angle $\phi_i(\Theta)$ in (26) can be rewritten as

$$\begin{cases} \phi_1(\Theta) = \arctan \dfrac{D_b - D_a + \sqrt{(D_b - D_a)^2 + 4D_{ab}^2}}{2D_{ab}} \\ \phi_2(\Theta) = \arctan \dfrac{D_b - D_a - \sqrt{(D_b - D_a)^2 + 4D_{ab}^2}}{2D_{ab}}. \end{cases} \tag{47}$$

First, the partial derivative of an arbitrary variable can be expressed as

$$\frac{\partial J(\Theta)}{\partial \varsigma} = \frac{\partial J(\Theta)}{\partial D_a}\frac{\partial D_a}{\partial \varsigma} + \frac{\partial J(\Theta)}{\partial D_b}\frac{\partial D_b}{\partial \varsigma} + \frac{\partial J(\Theta)}{\partial D_{ab}}\frac{\partial D_{ab}}{\partial \varsigma} \tag{48}$$

where $\varsigma$ represents one of the five variables $[\theta, \varphi, \gamma, u_0, v_0]^T$. The partial derivative of $D_a$ can be expressed as (49), show at the bottom of the page, $(\partial|\phi_i(\Theta) - \tilde{\phi}_i|/\partial D_a)$ can be expressed as

$$\frac{\partial\left|\phi_i(\Theta) - \tilde{\phi}_i\right|}{\partial D_a} = \frac{\phi_i(\Theta) - \tilde{\phi}_i}{\left|\phi_i(\Theta) - \tilde{\phi}_i\right|}\frac{\partial \phi_i(\Theta)}{\partial D_a}. \tag{50}$$

$(\partial\phi_1(\Theta)/\partial D_a)$ and $(\partial\phi_2(\Theta)/\partial D_a)$ are given as follows:

$$\begin{cases} \dfrac{\partial \phi_1(\Theta)}{\partial D_a} = \dfrac{-2D_{ab} + \frac{2D_{ab}(D_a - D_b)}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}}}{\left(D_b - D_a + \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)^2 + 4D_{ab}^2} \\ \dfrac{\partial \phi_2(\Theta)}{\partial D_a} = \dfrac{-2D_{ab} - \frac{2D_{ab}(D_a - D_b)}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}}}{\left(D_b - D_a - \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)^2 + 4D_{ab}^2}. \end{cases} \tag{51}$$

Simplifying (51) yields the following equation:

$$\frac{\partial \phi_1(\Theta)}{\partial D_a} = \frac{\partial \phi_2(\Theta)}{\partial D_a} = \frac{-D_{ab}}{4D_{ab}^2 + (D_b - D_a)^2}. \tag{52}$$

In (49), $(\partial|\lambda_i(\Theta) - \tilde{\lambda}_i|/\partial D_a)$ can be expressed as

$$\frac{\partial\left|\lambda_i(\Theta) - \tilde{\lambda}_i\right|}{\partial D_a} = \frac{\lambda_i(\Theta) - \tilde{\lambda}_i}{\left|\lambda_i(\Theta) - \tilde{\lambda}_i\right|} \cdot \frac{\partial \lambda_i(\Theta)}{\partial D_a}. \tag{53}$$

$(\partial\lambda_1(\Theta)/\partial D_a)$ and $(\partial\lambda_2(\Theta)/\partial D_a)$ are given as follows:

$$\begin{cases} \dfrac{\partial \lambda_1(\Theta)}{\partial D_a} = \dfrac{1}{2} + \dfrac{D_a - D_b}{2\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}} \\ \dfrac{\partial \lambda_2(\Theta)}{\partial D_a} = \dfrac{1}{2} - \dfrac{D_a - D_b}{2\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}}. \end{cases} \tag{54}$$

Referring to (49)–(54), $(\partial\phi_i(\Theta)/\partial D_b)$, $(\partial\phi_i(\Theta)/\partial D_{ab})$, $(\partial\lambda_i(\Theta)/\partial D_b)$, and $(\partial\lambda_i(\Theta)/\partial D_{ab})$ can be expressed as follows.

Simplifying (55), as shown at the bottom of the next page, yields (56), as shown at the bottom of the next page.

$$\frac{\partial J(\Theta)}{\partial D_a} = \sum_{m=1}^{M}\left\{\omega_1 \cdot \sum_{i=1}^{2}\frac{\partial\left|\phi_i(\Theta) - \tilde{\phi}_i\right|}{\partial D_a} + \omega_2 \cdot \sum_{i=1}^{2}\frac{\partial\left|\lambda_i(\Theta) - \tilde{\lambda}_i\right|}{\partial D_a}\right\} \tag{49}$$

Simplifying (57), as shown at the bottom of the page, yields the following equation:

$$\frac{\partial\phi_1(\Theta)}{\partial D_{ab}} = \frac{\partial\phi_2(\Theta)}{\partial D_{ab}} = \frac{D_a - D_b}{4D_{ab}^2 + (D_b - D_a)^2} \tag{58}$$

$$\begin{cases} \dfrac{\partial\lambda_1(\Theta)}{\partial D_b} = \dfrac{1}{2} + \dfrac{D_b - D_a}{2\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}} \\ [1.3pc]\dfrac{\partial\lambda_2(\Theta)}{\partial D_b} = \dfrac{1}{2} - \dfrac{D_b - D_a}{2\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}} \end{cases} \tag{59}$$

$$\begin{cases} \dfrac{\partial\lambda_1(\Theta)}{\partial D_{ab}} = \dfrac{2D_{ab}}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}} \\ [1.3pc]\dfrac{\partial\lambda_2(\Theta)}{\partial D_{ab}} = -\dfrac{2D_{ab}}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}}. \end{cases} \tag{60}$$

As shown in (48), the partial derivatives of the five variables differ from each other in $(\partial D_a/\partial\varsigma)$, $(\partial D_b/\partial\varsigma)$, and $(\partial D_{ab}/\partial\varsigma)$, where $\varsigma$ represents the five variables. Combined with (12)–(16) and (43), $(\partial D_a/\partial\varsigma)$, $(\partial D_b/\partial\varsigma)$, and $(\partial D_{ab}/\partial\varsigma)$ can be expressed as follows:

$$\frac{\partial D_a}{\partial\Phi} = \frac{2\tilde{r}_1 u_0^2}{3}\frac{\partial\tilde{r}_1}{\partial\Phi} + \frac{2\tilde{r}_2 v_0^2}{3}\frac{\partial\tilde{r}_2}{\partial\Phi} \tag{61}$$

where $\Phi$ denotes the angle variables $(\alpha, \beta, \gamma)$ and $\partial\tilde{r}_1/\partial\Phi$ and $\partial\tilde{r}_2/\partial\Phi$ can be expressed as follows:

$$\begin{cases} \dfrac{\partial\tilde{r}_1}{\partial\Phi} = \mathbf{E}_1\mathbf{r}^T\dfrac{\partial\mathbf{R}}{\partial\Phi} \\ \dfrac{\partial\tilde{r}_2}{\partial\Phi} = \mathbf{E}_2\mathbf{r}^T\dfrac{\partial\mathbf{R}}{\partial\Phi} \end{cases} \tag{62}$$

where $\mathbf{E}_1 = [1, 0, 0]$ and $\mathbf{E}_2 = [0, 1, 0]$. $\partial\mathbf{R}/\partial\Phi$ is given as follows:

$$\begin{cases} \dfrac{\partial\mathbf{R}}{\partial\theta} = \dfrac{\partial\mathbf{R}_z(\theta)}{\partial\theta}\mathbf{R}_y(\varphi)\mathbf{R}_x(\gamma) \\ \dfrac{\partial\mathbf{R}}{\partial\varphi} = \mathbf{R}_z(\theta)\dfrac{\partial\mathbf{R}_y(\varphi)}{\partial\varphi}\mathbf{R}_x(\gamma) \\ \dfrac{\partial\mathbf{R}}{\partial\gamma} = \mathbf{R}_z(\theta)\mathbf{R}_y(\varphi)\dfrac{\partial\mathbf{R}_x(\gamma)}{\partial\gamma} \end{cases} \tag{63}$$

where

$$\frac{\partial\mathbf{R}_z(\theta)}{\partial\theta} = \begin{bmatrix} -\sin\theta & -\cos\theta & 0 \\ \cos\theta & -\sin\theta & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{64}$$

$$\frac{\partial\mathbf{R}_y(\varphi)}{\partial\varphi} = \begin{bmatrix} -\sin\varphi & 0 & \cos\varphi \\ 0 & 0 & 0 \\ -\cos\varphi & 0 & -\sin\varphi \end{bmatrix} \tag{65}$$

$$\frac{\partial\mathbf{R}_x(\gamma)}{\partial\gamma} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & -\sin\gamma & -\cos\gamma \\ 0 & \cos\gamma & -\sin\gamma \end{bmatrix}. \tag{66}$$

The expressions for $(\partial D_a/\partial u_0)$ and $(\partial D_a/\partial v_0)$ are given as follows:

$$\begin{cases} \dfrac{\partial D_a}{\partial u_0} = \dfrac{2\tilde{r}_1^2 u_0}{3} \\ \dfrac{\partial D_a}{\partial v_0} = \dfrac{2\tilde{r}_2^2 v_0}{3}. \end{cases} \tag{67}$$

Similar to (61) and (67), one can derive the expression for $(\partial D_b/\partial\varsigma)$ and $(\partial D_{ab}/\partial\varsigma)$ as follows:

$$\begin{cases} \dfrac{\partial D_b}{\partial\Phi} = \dfrac{2\tilde{f}_1 u_0^2}{3}\dfrac{\partial\tilde{f}_1}{\partial\Phi} + \dfrac{2\tilde{f}_2 v_0^2}{3}\dfrac{\partial\tilde{f}_2}{\partial\Phi} \\ \dfrac{\partial D_{ab}}{\partial\Phi} = \dfrac{u_0^2}{3}\left(\dfrac{\partial\tilde{r}_1}{\partial\Phi}\tilde{f}_1 + \tilde{r}_1\dfrac{\partial\tilde{f}_1}{\partial\Phi}\right) + \dfrac{v_0^2}{3}\left(\dfrac{\partial\tilde{r}_2}{\partial\Phi}\tilde{f}_2 + \tilde{r}_2\dfrac{\partial\tilde{f}_2}{\partial\Phi}\right) \end{cases} \tag{68}$$

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$$\begin{cases} \dfrac{\partial\phi_1(\Theta)}{\partial D_b} = \dfrac{2D_{ab} + \frac{2D_{ab}(D_b - D_a)}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}}}{\left(D_b - D_a + \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)^2 + 4D_{ab}^2} \\ \dfrac{\partial\phi_2(\Theta)}{\partial D_b} = \dfrac{2D_{ab} - \frac{2D_{ab}(D_b - D_a)}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}}}{\left(D_b - D_a - \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)^2 + 4D_{ab}^2} \end{cases} \tag{55}$$

$$\frac{\partial\phi_1(\Theta)}{\partial D_b} = \frac{\partial\phi_2(\Theta)}{\partial D_b} = \frac{D_{ab}}{4D_{ab}^2 + (D_b - D_a)^2} \tag{56}$$

$$\begin{cases} \dfrac{\partial\phi_1(\Theta)}{\partial D_{ab}} = \dfrac{\frac{8D_{ab}{}^2}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}} - 2\left(D_b - D_a + \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)}{\left(D_b - D_a + \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)^2 + 4D_{ab}^2} \\ \dfrac{\partial\phi_2(\Theta)}{\partial D_{ab}} = \dfrac{-\frac{8D_{ab}{}^2}{\sqrt{(D_a - D_b)^2 + 4D_{ab}^2}} + 2\left(D_b - D_a - \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)}{\left(D_b - D_a - \sqrt{(D_a - D_b)^2 + 4D_{ab}^2}\right)^2 + 4D_{ab}^2} \end{cases} \tag{57}$$

$$\begin{cases} \dfrac{\partial D_b}{\partial u_0} = \dfrac{2\tilde{f}_1^2 u_0}{3}, & \dfrac{\partial D_b}{\partial v_0} = \dfrac{2\tilde{f}_2^2 v_0}{3} \\ \dfrac{\partial D_{ab}}{\partial u_0} = \dfrac{2\tilde{r}_1 \tilde{f}_1 u_0}{3}, & \dfrac{\partial D_{ab}}{\partial v_0} = \dfrac{2\tilde{r}_2 \tilde{f}_2 v_0}{3} \end{cases} \tag{69}$$

where

$$\begin{cases} \dfrac{\partial \tilde{f}_1}{\partial \Phi} = \mathbf{E}_1 \mathbf{f}_a^T \dfrac{\partial \mathbf{R}}{\partial \Phi} \\ \dfrac{\partial \tilde{f}_2}{\partial \Phi} = \mathbf{E}_2 \mathbf{f}_a^T \dfrac{\partial \mathbf{R}}{\partial \Phi}. \end{cases} \tag{70}$$

With (61)–(70), the gradient of loss function $J(\Theta)$ can be obtained as shown in (29).

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